



Benefit-Underutilization Analytics

Overview

Card issuers struggle to measure how much benefit value goes unclaimed, making it difficult to justify benefit investments or prove ROI to stakeholders. Card members, meanwhile, are often unaware of what they are missing, reducing their perceived card value and engagement. This challenge asks you to build an analytics engine that quantifies, in dollar terms, the benefit value each card member leaves on the table across unused credits, lounge visits, and protection benefits, and then turns that gap into a personalized, measurable engagement opportunity. Surfacing this value proactively can drive meaningful improvements in retention and satisfaction.

Tech stack

- Open to all
- Examples for your reference only
 - Frontend: React, Tableau Embedded
 - Backend & APIs: Python (FastAPI), Node.js
 - Data analytics tools: Python (pandas, NumPy), R
 - Machine learning frameworks: scikit-learn, XGBoost
 - Cloud computing platforms: AWS, GCP
 - Database management systems: Snowflake, BigQuery

Tasks

- Design an algorithm to map each card member's transaction history to their full set of benefit entitlements.
- Calculate the dollar value of unclaimed benefits, covering unused credits, lounge visits, and protection coverage per card member.
- Develop a dashboard or customer-facing interface that surfaces the top unclaimed benefits in clear, personalized terms.

- Implement nudge logic to trigger targeted engagement prompts at the right moment based on spending patterns.
- Test and optimize the analytics engine for entitlement mapping accuracy, nudge relevance, and measurable uplift in benefit engagement.

Submission format

- Project description (mandatory)
- Presentation (mandatory)
- Any other documentation or file related to the proposed solution
- Video and link of the project

Resources

- [XGBoost Documentation](#)
- [Google BigQuery](#)
- [FastAPI Documentation](#)

Cross-Channel Journey Stitching



Overview

If customer data is siloed across channels (app, website, and phone), it makes it nearly impossible to understand the full customer journey, identify where experiences break down, or intervene before a customer churns or escalates. This challenge asks you to build a cross-channel identity resolution and event-stitching platform that assembles a unified timeline per customer across app, web, call-center, and in-person touchpoints. Unified journey visibility enables teams to find and fix the exact moments where the experience fails.

Tech stack

- Open to all